SAR Change Detection — Riyadh Urban Development (2022–2024)

Ahmed | Electrical Engineer | Earth Observation Portfolio | April 2026

Objective

Detect and classify urban change across Riyadh, Saudi Arabia between January 2022 and February 2024 using Sentinel-1 Synthetic Aperture Radar (SAR) imagery. The goal is to demonstrate an end-to-end SAR processing pipeline — from raw data ingestion to machine-learning classification — producing actionable change maps for urban monitoring applications.

Data

Parameter	Value
Satellite	Sentinel-1A (ESA Copernicus)
Mode / Product	Interferometric Wide (IW) / GRD-COG
Polarisation	VV (co-polarised)
Before scene	25 Jan 2022 (orbit 41619, rel. orbit 72, ascending)
After scene	20 Feb 2024 (orbit 52644, rel. orbit 72, ascending)
AOI	Riyadh: 24.55–24.85°N, 46.55–46.95°E
Pixel resolution	10 m (UTM Zone 38N, EPSG:32638)
Data source	Copernicus Dataspace (OData API)

Methodology

- **Georeferencing:** Parsed 210 geolocation grid points (GCPs) from Sentinel-1 annotation XML files and assigned EPSG:4326 CRS to the measurement TIFFs, which ship with CRS=None in COG format.
- **Reprojection & alignment:** Warped both scenes via GCP-based transformation onto a common 10 m UTM grid (EPSG:32638) covering the Riyadh AOI. Verified pixel-to-pixel alignment (identical shapes and transforms).
- **Speckle filtering:** Applied a Lee filter (7×7 window) on linear amplitude to reduce multiplicative SAR noise while preserving edges, then converted to decibels (dB).
- **Change detection:** Computed log-ratio difference ($dB_{2024} - dB_{2022}$); classified pixels exceeding ±3 dB as significant change (≈ factor-of-2 change in backscatter power).
- **ML classification:** Built a 4-band feature stack (dB 2022, dB 2024, |change|, signed change) and applied K-Means (k=5) and Gaussian Mixture Models. K-Means achieved a silhouette score of 0.319. Clusters were automatically interpreted as: stable urban, stable desert, new construction, land clearing, and seasonal variation.

Key Result

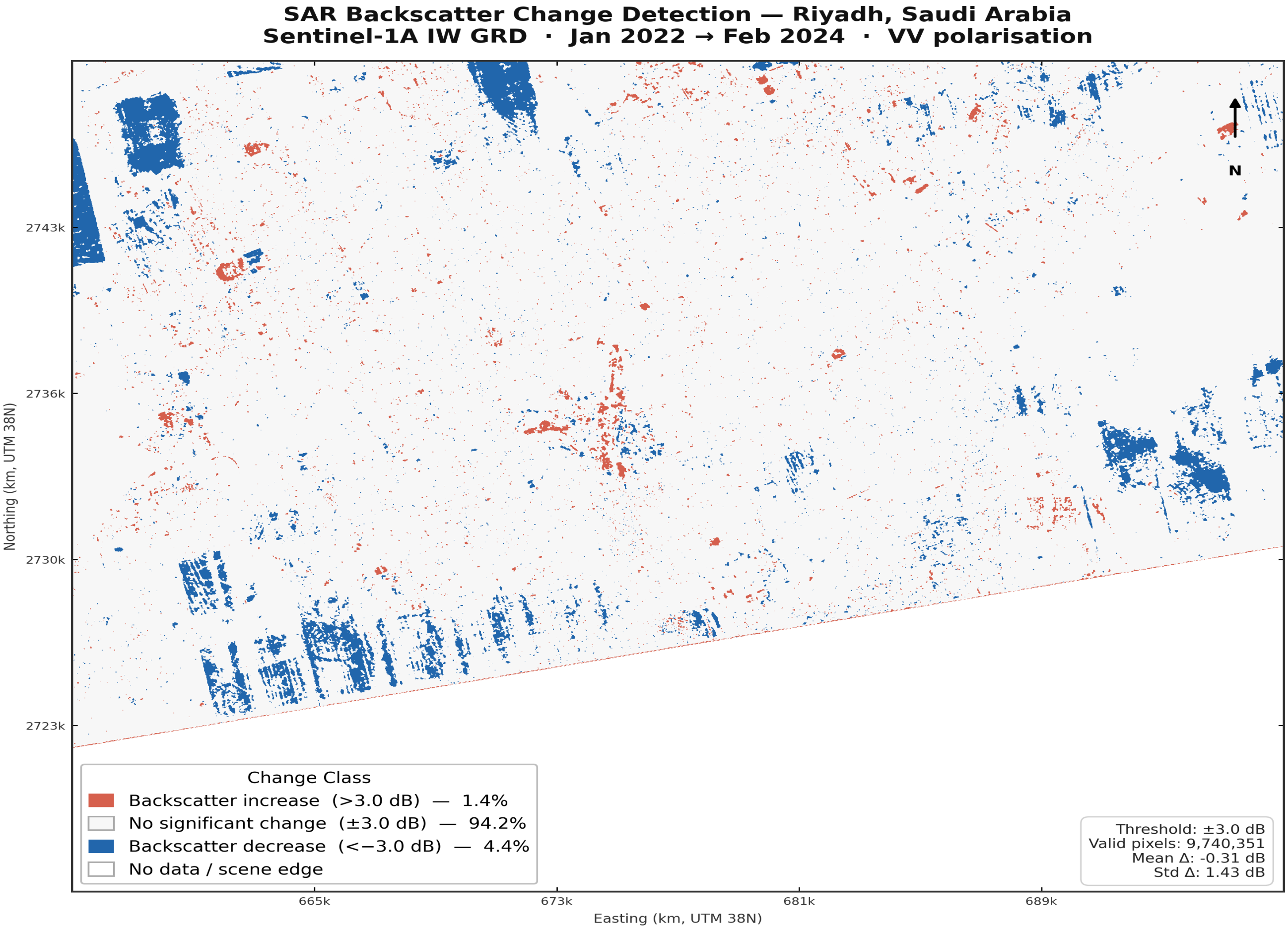

Figure 1. SAR backscatter change map (±3 dB threshold). Red = increased backscatter (new construction); Blue = decreased backscatter (clearing/demolition). 94.2% stable, 1.4% increase, 4.4% decrease across 9.7M valid pixels.

Validation

Connected-component analysis identified the top 10 change clusters. Cross-referencing with Google Earth optical imagery confirmed:

- **King Salman Park** (24.83°N, 46.58°E): 496 ha decrease — massive clearing for the 13.4 km² mega-project, clearly visible in optical imagery as demolished blocks.
- **Northern expansion** (24.84°N, 46.69°E): 383 ha decrease — new neighbourhood grading and road infrastructure preparation.
- **New residential construction** (24.72°N, 46.71°E): 33.5 ha increase — new buildings producing higher radar returns where open land previously existed.

Conclusion

This project demonstrates that freely available Sentinel-1 SAR data, combined with proper geocorrection and standard image processing techniques, can reliably detect and classify urban change at 10 m resolution. The pipeline runs end-to-end in Python, requires no commercial software, and produces GIS-ready outputs. SAR's all-weather, day/night imaging capability makes it particularly valuable for continuous urban monitoring in arid regions like Riyadh, where cloud cover is rare but optical revisit may still be limited. The validated results confirm detection of known mega-projects (King Salman Park) and ongoing suburban expansion, demonstrating practical applicability for urban planning, infrastructure monitoring, and development tracking.